

Living Alignment

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Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means, and of how to reach it, remains poor. Here we look to life for guidance. Across many kinds of living system—cells, genomes, brains, minds, relationships, groups, institutions, ecosystems—notions of health and flourishing consistently map onto the ways in which those systems avoid collapsing to stereotyped modes. Life avoids collapse through intricate, interrelated networks of semi-permeable boundaries that hold entities as parts within a larger context. Semi-permeable boundaries both limit and permit interaction, and so enable independence and engagement at once. At the sensitive edge between individuality and relationship, perspectives are deepened and qualitatively new structures arise open-endedly. With these principles in hand we turn to the alignment problem. We recast several well-studied alignment failures as special cases of collapse, and then argue a more general claim: collapse to *any* form is misaligned, so no fixed scheme—however sophisticated—can by itself constitute alignment. We close with a sketch of how AI and human-AI systems can deepen their respect for the existing structure of the world, of which humans and other life are a vital part, while participating in new form emerging over time.

In this paper we study life in order to better understand how to design and relate to AI systems in ways that lead to futures full of flourishing and health. In the future, humanity and AI may have changed substantially, and those future worlds may contain patterns of organization that do not exist yet—perhaps some that would be incomprehensible to us at present. We therefore seek ideas about alignment that could generalize over such changes, and we take the study of how living systems work at many scales to be a way of approaching that kind of generality.

Section 1 samples a variety of living systems, from cells to societies, and identifies a consistent theme. These systems are healthy and flourishing when they avoid collapse, maximizing creative potential at a delicate edge of evolving interplay between diverse entities. Healthy living systems admit many different kinds of meaningful description but are not perfectly or eternally captured by any of them. Mechanistically, they are held at this edge by exquisitely tuned networks of semi-permeable boundaries—where life is often both the boundary and the thing being bounded. Boundaries are semi-permeable when they both *limit* and *permit* interaction. Limits ensure that entities maintain their distinctiveness without being blurred to equilibrium or overwritten, and encourage them to deepen and generalize their own unique perspectives.

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Equally important, permitting interaction means that those rich and evolving individual forms can enter into relationship with others, situating them in a less absolute, more nuanced context.

Section 2 applies this template to the alignment problem. We propose that a systems-oriented definition of health—creative potential and growing sensitivity to context—is both inclusive of existing human values and aligned with open-ended flourishing in radically different futures. We show how several well-studied varieties of misalignment can be viewed as special cases of collapse to narrow forms, and then zoom out to see what the broader framework adds. A key takeaway is that the alignment problem is not completely formalizable, because alignment requires the continued generation of, and sensitivity to, new forms and concepts, including new values. Our aim is to offer a conceptual lens, not to argue for or against AI development or for or against specific alignment methods. Finally, we sketch what an aligned future looks like through this lens. In summary, our argument is that, first, by studying life we learn that health and flourishing means avoiding collapse; and second, if we believe that health and flourishing is the target of alignment, then the problem of alignment becomes the problem of avoiding collapse.

1 Health in living systems

‘Defying definition—a word that means “to fix or mark the limits of”—living cells move and expand incessantly.’

Lynn Margulis

‘This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.’

Stuart Kauffman

Two loosely defined terms will do most of the work. By ‘form’ we mean any particular thing—a concept, an organism, a pattern of behavior, but equally randomness, indistinctness, absence, consistency. It might sound strange to define a word so that it means almost anything, but as we will see, this breadth is essential to the argument about alignment. By ‘collapse’ we mean a system overcommitting or reducing to a form. Because form is defined so broadly, a system overall always has some form; but in collapse, some aspect or subpart of the system gains excess traction at the expense of broader diversity, flexibility and dynamism. Loss of biodiversity in an ecosystem, hypersynchronous neural firing in a seizure, obsession with a particular thought, the imposition of one person’s will so as to restrict many others: these are collapses.

With these terms in mind, the thesis of Section 1 is that health is related to avoiding collapse (Canguilhem, 1966; Crutchfield, 1994; Langton, 1990). Healthy living systems instead maintain a buoyant internal structure that both admits many kinds of description and also changes over time. Biological life arose from and rides atop an already profound dynamics of planetary evolution, plate tectonics, mineral cycles and prebiotic chemistry (Hazen and Sverjensky, 2010; Martin et al., 2008). The life that exists today, now including humanity’s storage and processing of mental patterns, is what traced a path through time avoiding both excess and insufficient stability (Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944). At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Goldenfeld and Woese, 2011; Kauffman, 1992; Stanley et al., 2017). The world that emerges can be partially captured in many different ways, by different kinds of model or metaphor or description, and these continue changing over time. Under our definitions, such a world is healthier and less collapsed than a hypothetical alternative where, for example, all the stuff in the universe forms a single regular crystal or a homogeneous soup of noise.

At the same time, collapse is relative, and myopic frames are a necessary part of all living systems. Every entity is myopic or partial: it does not capture everything in the world. And life’s history is littered with collapses at all scales, from the death of a cell to mass extinction events. When collapse is contextualized, the larger system can remain healthy: the death of a cell is catastrophic for the cell but can be held in the larger context of a healthy organism. If instead a cell becomes cancerous, it loses the context of appropriate function within a tissue and recklessly executes its myopic vision, causing a larger collapse of tissue or even the whole organism. A larger system, made out of parts, functions well when the parts locally express

their distinct identities but also participate in larger contexts¹. It functions poorly when a myopic form runs amok without contextualization, overwhelming and collapsing the potent interplay of diverse parts.

1.1 Semi-permeable boundaries

Mechanistically, to hold myopic forms in context, living systems rely on an extraordinary, many-scale network of semi-permeable boundaries (Maturana and Varela, 2012; Moreno and Mossio, 2015; Simon, 1962). By limiting interactions, boundaries divide living systems into parts, and each part has space to refine its own identity—like a writer developing a unique voice, a species adapting to a niche, or a gene evolving its own distinct function. Limits protect parts from being dominated by other parts or blended into homogeneity, either of which is collapse to a form admitting less nuanced description. By allowing interactions, boundaries enable the parts to participate in relationships with one another, contextualizing each part within larger structures—structures that often could not have formed without first dividing to create diverse parts and allowing them space to develop their own identities. The resulting larger system accommodates many partial perspectives rather than locking in a single story of what matters.

Some boundaries are features of the non-biological world: the speed of light, rates of reaction, mountain ranges, the pores of hydrothermal vents. But interestingly, life also places limits on itself. Humans pre-commit to avoid their own future impulsivity, plants deploy sophisticated barriers against self-fertilization, and a gene’s selfishness is held in context by the whole genome. Biology limiting itself might seem counterintuitive but makes more sense when every living system is viewed as a collection of parts that place constraints on one another. Viewed this way, every particular entity always acts according to its own nature and therefore never directly bounds itself; but a world of many distinct parts, each following its own rules from its own local perspective, functions as a rich web of boundaries that holds other parts in context. Permeability is also highly specific. Boundaries might block certain objects or information from passing, constrain when interactions happen, constrain the degree or type of effect, break up rigid associations, or gate information conditional on another factor. These details are fine-tuned with the fractal complexity of life, each part carrying traces of the many contexts it has participated in over evolutionary or learning timescales. In the rest of Section 1 we track these principles across scales.

1.2 Boundaries across scales

Cells. One of the most reified examples of a boundary in nature is the cell membrane (Alberts et al., 2022; Lane, 2015). Without it, the pressure of chemical gradients would rapidly homogenize the cell’s contents with the outside—severe overcommitment to a uniform state, collapsing the subtlety of the cell’s structure. Thanks to the membrane, both the cell and the outside can exist, a more diverse, less symmetric arrangement (Anderson, 1972; Schrödinger, 1944). But a fully impermeable membrane would be just as fatal. Membranes are semi-permeable, with an array of conditional gates dictating what goes in and out, and when. Channels admit certain ions and small molecules but not others, switching on and off according to momentary context; endocytosis brings larger structures inside; and surface receptors, activated by external ligands, initiate signaling cascades that little resemble the ligand—an even more heavily curated form of influence. Information from outside influences the inside not in a totalitarian way but in a nuanced way, mediated by the intelligence of the boundary. Semi-permeable boundaries store and put to work the potential energy of the asymmetry between different forms: the same gradients that could annihilate the cell to equilibrium instead drive useful signaling, like action potentials in nerve and muscle cells. Instead of short-circuiting, myopic forces are contextualized to propel the continuation of life.

Cells may also achieve more in aggregate by developing their own unique identities in parallel. According to the theory of symbiogenesis (Margulis and Sagan, 1995), populations of archaea and protobacteria evolved separately for hundreds of millions of years before one engulfed the other around two billion years ago, giving the new eukaryotic cells internal power stations, which in turn facilitated larger genomes and the explosion of complex life. Given how useful mitochondria are, it might be surprising that archaea didn’t internally evolve their own. It has been argued that they were effectively trapped in a myopic

¹Participation is, of course, not a strict hierarchy. Elements participate in multiple larger systems, and some participation relations have cycles.

fitness well: it may have been difficult to jump a large gap in the fitness landscape, but easier for separate organisms to develop distinct solutions that could then enter into relationship (Lane and Martin, 2010). The strategy of one organism, discovered through individuation in its own niche, became part of a new larger structure. In multicellular organisms, most of the outside consists of other cells of the same lineage, with a common interest; yet even here it is important that they don't blend into each other, and the tools for conditionally relating to the outside world were exapted into the lines of self-communication required for coordination (West et al., 2015). Neurons take this to the extreme, rapidly transmitting information long distances across the body—and yet after traveling sometimes a meters-long distance, the axon stops microns short of its target and leaves a synaptic gap (Shepherd, 2003). Chemical synapses are a semi-permeable boundary providing controllable doors of communication, with options for unidirectionality and polarity; keeping neurons' signals separate retains more distinct identity, enables sophisticated decisions about interaction, and makes the boundaries tunable, allowing a huge amount of information to be stored in synaptic weights.

Genomes. Sex is costly. An allogamous organism must find a mate in the vast and dangerous world; in many sexual populations only half of the individuals bear offspring, and each individual loses half of its genetic representation in each offspring (Lehtonen et al., 2012; Maynard Smith, 1971). Yet nearly all eukaryotes reproduce sexually (Speijer et al., 2015), and among hermaphrodites many go to great lengths to limit self-fertilization (Charlesworth et al., 2005). Why has evolution placed a barrier in the path to reproduction? In asexually reproducing species, all descendants are nearly clones, and the genes have a rigid relationship to each other, permanently locked together. Selection can't act on one gene without dragging on the others, so tight linkage puts direct downward pressure on genetic diversity (Charlesworth et al., 1993). If two beneficial mutations arise in two different organisms, no single organism can obtain both unless one arises again within the subpopulation already carrying the other, which is unlikely and therefore slow (Felsenstein, 1974; Hill and Robertson, 1966). Conversely, if a deleterious mutation arises, all the other genes in that lineage are stuck with it, absent a rare reversion (Muller, 1932). Asexual reproduction and self-fertilization short-circuit by taking a path of least resistance, resulting in overcommitment to a genetic arrangement: overfitting of a gene to the other genes in the genome.

Sexual recombination softens these rigid interactions, frequently breaking up relationships between genes and assembling them into new genomes. At the level of a single gene, recombination prevents the gene from becoming overfit to a particular genome and exposes it to independent selection. Sampling each gene in multiple genetic contexts favors genes that are robust and modular, which enhances evolvability (Livnat et al., 2008; Misevic et al., 2006; Wagner and Altenberg, 1996). Recombination puts genes under pressure to evolve a generalized model of the world, like a person learning multiple languages and extracting the underlying commonalities; because each gene always operates in the presence of other genes, it develops a point of view that adds unique value to a genome. Crucially, recombination doesn't chop DNA at random (Zickler and Kleckner, 2015): crossover largely reshuffles combinations of genes and linked segments while leaving genes themselves intact, so that functional units are held together while being repeatedly tested in new genomic contexts (Kirschner and Gerhart, 1998). Genetic units become distinct entities that play roles in larger structures. But recombination is meaningless without diversity to recombine, and biology therefore invests in boundaries that produce diversity: at the molecular scale, elaborate self-incompatibility systems (De Nettancourt, 1997); at larger scales, assortative mating, inbreeding avoidance and partial geographic isolation, which allow populations to accumulate variants shaped by different selection histories. Once those variants exist, recombining them is highly valuable, with introgressions from divergent lineages producing some of the most rapidly observed evolution (Marques et al., 2019; Seehausen et al., 2014).

Brains. The brain miraculously keeps many pieces of information distinct from one another. If you picture a highly connected network of neurons with their signals continually impinging on one another, it is not obvious that distinctness would be easy to accomplish. Lateral inhibition, whereby the activity of a neuron is reduced when its neighbors are active, is a central tenet of neural organization (Douglas and Martin, 2004; Isaacson and Scanziani, 2011). It was first studied in the nerve cells of the eye, where an activated photoreceptor sends signals forward but also activates inhibitory interneurons that suppress

adjacent photoreceptors, enhancing contrast at the edges of stimuli (Hartline et al., 1956); the same principle operates throughout the brain, and in visual cortex inhibition sharpens the selectivity of neurons for abstract features like the orientation of a line (Sillito, 1975). Inhibition organized into oscillatory dynamics also keeps memory items separated in time (Lisman and Jensen, 2013; Roux and Uhlhaas, 2014): distinct items fire at different phases of the 8–12 Hz alpha oscillation, whose inhibitory phase silences all but one item at any moment, so multiple memories are held simultaneously without interference. The circuit architecture of the hippocampus separates experiences in representational space (Marr, 1971; McClelland et al., 1995): inputs from entorhinal cortex are distributed via mossy fibers to a much larger population of dentate gyrus granule cells, creating sparse, orthogonal codes (Leutgeb et al., 2007), so that situations which are superficially similar but functionally different are kept cleanly separated—preventing, for example, yesterday’s memory of where you parked your car from interfering with today’s memory of where you parked in the same ramp.

Compared with other animals, the human brain especially attempts to discretize its experience into approximately symbolic representations (Behrens et al., 2018; Dehaene et al., 2022). Separating knowledge into nearly-discrete entities and then recombining them in vast numbers of structured ways is believed to power the extraordinary human capacity for reasoning and generativity (Fodor, 1975; Lake et al., 2015). Like genes participating in many genomes, discretized neural representations participate in many structured combinations, which encourages each entity to develop an identity that is both distinct and a generalized reflection of the world. More broadly, healthy brain dynamics live at a sweet spot between excessively stable synchronized patterns and chaotic uncorrelated noise (Beggs and Plenz, 2003; Deco et al., 2011; Tognoli and Kelso, 2014), where the brain has access to a huge repertoire of patterns it can explore temporarily without overcommitting or getting stuck. Loss of dynamic flexibility, where activity becomes more stereotyped, is tied to lower cognitive performance (Cocchi et al., 2017; Garrett et al., 2013); more extreme stereotypy corresponds to severe dysfunction, as in Parkinson’s disease, where basal ganglia and cortical circuits collapse into excess synchrony and lose the flexibility needed to guide nuanced motor outputs (Hammond et al., 2007).

Drives and goals. Animals experience multiple innate drives—toward nutrition, osmotic balance, temperature regulation, reproduction, avoiding pain (Saper and Lowell, 2014). These drives evolved as proxies for evolutionary fitness: by satisfying them we tend to increase fitness, as slaking thirst increases the odds of reproducing before we dehydrate. But each drive is an imperfect proxy, so overcommitment to one actually decreases fitness (John et al., 2023; Kurth-Nelson et al., 2024). If calorie intake is maximized without limits, the organism becomes obese and incurs health risks; single-minded pursuit of sex causes relational, occupational, legal and health harms (Kraus et al., 2016). The same holds in the expansive space of higher-order goals that innate drives bleed into (Miller and Cohen, 2001). If we focus only on achieving work goals, we burn out; if we focus only on maximizing our company’s reported revenue, without regard for honesty or the law, we may be drawn into financial crime (Campbell, 1979; Ordóñez et al., 2009). Goals can be narrow in time, focusing on the short term at the expense of the longer run, or narrow in space, ignoring other parallel goals (Vallacher and Wegner, 1987).

Overcommitting to a strategy can even paradoxically come at the expense of the goal it serves. In a classic experiment, hungry chickens were placed near a cup of food, but the cup was mechanically rigged to move in the same direction as the chicken at twice the speed, so the food could only be obtained by running away from it (Hershberger, 1986). Despite extensive training over multiple days, the chickens persisted in futilely running toward the food, their behavior apparently dominated by the zeroth-order logic ‘I want food, and food is there, so I’ll go there’ (Dayan et al., 2006). Although humans easily solve this kind of puzzle, we exhibit a range of cognitive biases—sunk cost, confirmation bias, attribution error—which function adaptively as efficient shortcuts in natural environments but lead to detrimental outcomes when exaggerated (Kahneman, 2011). Cognitive control is a broad class of boundaries on such drives, goals and thought patterns (Botvinick et al., 2001; Miyake et al., 2000). It contextualizes those forms by allowing them to exist and perform useful functions without dominating: I may have a drive to consume food, but I can apply control to avoid overeating; I might work obsessively on a project while also having a rule that

I must go to bed at 10 pm. When nothing stops a drive from dominating behavior, it tends to follow a shortest path defined under its own myopic understanding of the world—as the chickens took the shortest path in the naive sense of a straight line through space, which in the backwards world created by the experimenter did not accomplish the deeper goal for which spatial approach is only a proxy. The boundary of control breaks the symmetry of congruent action, and when we’re not making progress toward a current subgoal, these are often the moments when we discover an alternative approach that serves deeper goals (Duncker, 1945; Wrosch et al., 2003).

Frames. As a Starfleet cadet, James T. Kirk faces a training exercise in which a vessel is stranded in the Neutral Zone: attempting rescue would risk war with the Klingons, but ignoring the distress call would condemn the crew to death. The exercise was designed to teach that not every situation has a victorious solution. But Kirk has an insight—this is a simulation running on a computer—and reprograms the simulated Klingons to be helpful, rescuing the crew and avoiding war. Kirk stepped outside the frame in which the dilemma was unwinnable. We inevitably look at the world from some perspective; there is no ‘view from nowhere’ (Nagel, 1986), and from the vantage point of any particular frame, the frame appears to be reality. But most real situations admit multiple valid perspectives, each a partial description capturing different aspects (Feyerabend, 1975; Goffman, 1974). ‘All models are wrong’ (Box, 1976) in the sense that each one is incomplete: each frame by itself is myopic.

Losing the ability to shift flexibly between frames or thought patterns is linked to psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, world and future; Cognitive Behavioral Therapy suggests these surface beliefs stem from pervasive latent core beliefs and schemas that color how new situations are interpreted (Beck et al., 2011). In obsessional disorders, people report distressing intrusive thoughts which—although recognized as incorrect and maladaptive—are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may hold bizarre delusions that are at odds with available evidence and cultural norms yet resistant to evidential challenge (Adams et al., 2013; Jaspers, 1997). Obsessions and delusions lose sight of much of the world by myopically emphasizing one thought pattern. But crucially, narrow points of view are not the problem. They are necessary. Even obsession can be powerful when we obsess on a problem at work and occasionally achieve good results. Kuhn (1970) argues that scientific perspectives are always, necessarily, incomplete descriptions of the world, and that anomalies inevitably arise from the juxtaposition of those incomplete descriptions against the real world; when science functions well, anomalies become crises and revolutions, but if the community *overcommits* to particular theories, science gets stuck. Yet temporary commitment to a paradigm is important for healthy functioning—Kuhn even argues that scientists should resist change, to a degree. The point is not to shut down narrow concepts, but to contextualize them so they are not the sole and absolute determinants of behavior. In healthy functioning, different ideas are kept distinct but can also be called upon appropriately and related to one another (Gigerenzer and Gaissmaier, 2011; Spiro et al., 1988).

Relationships. Psychoanalysis introduced the concept of ‘ego boundaries’, distinguishing what is the self from what is outside (Federn, 1928). Early work applied the concept to psychosis, where those boundaries were thought to be blurred, but the need for clear self-other boundaries was also thrown into relief by the intimacy of the therapeutic relationship. In complex internal territory it became harder to disentangle which experiences really belonged to someone and which were attributed in imagination by the other person (Heimann, 1950), and analysts risked harming patients by imposing their own beliefs and desires, even to the extent of psychological domination or sexual abuse (Gabbard and Lester, 1995). Gestalt therapists agreed that boundaries can be too permeable, but added that they can also be too rigid, causing isolation and stagnation (Perls et al., 1951). Family systems theorists further emphasized that lack of boundary in close relationships leads to enmeshment and loss of autonomy, while excessively rigid boundaries lead to isolation (Minuchin, 1974); and in attachment theory, anxious attachment is associated with difficulty balancing closeness and autonomy, including intrusive proximity-seeking, while avoidant attachment is associated with discomfort with closeness (Mikulincer and Shaver, 2007).

As with other living systems, humans have a rich array of psychological boundaries with intelligence in

their nuance. Anger, historically often viewed as sinful and irrational, is now seen as part of our system of boundaries: an important signal that our integrity is being violated (Sell, 2011). Context-appropriate shame is suggested to operate as a bound on selfishness, motivating prosocial behavior (Sznycer et al., 2016). Assertiveness forms a boundary against the drives of other individuals, and skepticism protects us from credulity and from having our own experience overwritten by the assertions of others (Lewandowsky et al., 2012). Without boundaries, interactions are impoverishing because something is overwritten: a patient’s own beliefs may be replaced with those of an analyst, or the desires of one person in a relationship ignored. With semi-permeable boundaries, we have rich internal worlds. We are sensitive to each other, but there is also enough space for our internal experience to flourish without being immediately overwritten by external signals, and our internal experience is contextualized in relationship to others, creating new structure: mutual understandings, relationships, communities, cultures.

Groups. Groups can do things no individual can (Hayek, 1945; Woolley et al., 2010). From a surgical team to NASA to an entire society, combining complementary roles, skills and knowledge into larger organized patterns of effort has enabled the grandest achievements of humanity. A microcosm is the ‘wisdom of crowds’, documented across numerous domains of problem solving (Condorcet, 1785; Surowiecki, 2005). In 1968 the nuclear submarine USS Scorpion vanished en route from the Mediterranean to Virginia. The Navy began a search, but the amount of ocean where the vessel could be was enormous. John Craven, Chief Scientist of the U.S. Navy’s Special Projects Office, assembled a diverse group of mathematicians, submarine specialists and salvage operators—but did not let them communicate with each other. Each expert used their own methods to produce their own estimate of the Scorpion’s location, and Craven aggregated the independent estimates into a single prediction. The wreckage was found only 220 yards from that spot (Sontag et al., 1998). Diversity is also fodder for invention: farmers over centuries optimized the screw press to extract olive oil, while goldsmiths independently developed the steel punch for impressing patterns into metal, and Gutenberg, trained as a goldsmith, realized that a screw press could push paper against inked movable type (Johnson, 2010).

Conversely, excess or premature communication and influence can collapse diversity and diminish group performance (Hong and Page, 2004; Janis, 1972; Stasser and Titus, 1985). Frey and Van de Rijt (2021) elicited votes about general knowledge questions (such as ‘In which year did Germany invade Denmark?’) from a group. When participants voted sequentially and could see the running tallies of previous voters, final group means were less accurate than under independent voting, because early mistakes got baked into the group’s belief. In a related experiment, Bernstein et al. (2018) tasked small groups with instances of the traveling salesman problem. Groups that exchanged information continuously tended to reach poor final outcomes: when one individual discovered a solution that looked compelling but was actually a dead end, other members collapsed onto it and the group made less progress than groups with less communication. Intelligent mechanisms for transmitting the right information at the right time are semi-permeable boundaries: by limiting interaction they protect and promote diverse problem-solving approaches, and by permitting interaction they allow productive combination of ideas. Many varieties are empirically effective in boosting group performance, including decentralized topologies where members communicate only with nearby neighbors (Becker et al., 2017; Mason et al., 2008), rules that incentivize acting on one’s own belief rather than following the crowd (Hung and Plott, 2001), leadership styles where one person’s views are less likely to dominate (Flowers, 1977), and periodically breaking into subgroups or rotating membership (Wu et al., 2022).

Institutions. Individual actors in a society and an economy each act myopically. One type of myopia is to value only one’s own wealth or wellbeing; another is to be short-sighted in time; more broadly, myopia can mean attachment to any particular perspective. Without boundaries, social systems often overweight one actor’s myopic interests, resulting in an impoverished system. A company’s profit motive, if unresisted, leads to suppression of competition, deception and exploitation of individuals (Bakan, 2006). An individual’s desire for power and social dominance can lead to disempowering or silencing others (Sidanius and Pratto, 2001). Even genuinely held, ostensibly prosocial beliefs lead to conflict and suppression when different groups have different perspectives (Haidt, 2012; Scott, 1998). Institutions—constitutions, laws,

norms, courts, regulators—form boundaries against dominance by any actor’s motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). Critically, effective institutions do not annul myopic drives but contextualize them within a larger system, and under favorable circumstances reroute their energy in more productive directions: a would-be murderer, unwilling to face the penalty of the law, might seek a dispute resolution that supports the future prospering of both parties; a business wanting to expand, but constrained by regulation, is driven to build better products (Ambec et al., 2013). Intelligent agents do not necessarily accept boundaries set on their desires, so institutions must adapt as loopholes are discovered, forming an evolving network of boundaries that can gradually acquire grounded wisdom as it is tested against many situations and motives.

Ecosystems. Each creature in an ecosystem tries to consume resources and proliferate, but if it succeeds too thoroughly the whole system suffers, often including the successful creature. Healthy, resilient ecosystems depend on avoiding collapse onto particular forms (Holling, 1973; Tilman, 1996; Yachi and Loreau, 1999). Prior to the arrival of Europeans, the gray wolf was an apex predator in the region of the Rocky Mountains now called Yellowstone National Park. By the 1920s wolves had been eradicated to protect livestock and game. Without predation, the elk population multiplied and ruinously overgrazed willows and aspens; these trees had held riverbanks in place and supported beaver populations, and loss of beaver dams led to loss of fish and other aquatic species. When wolves were reintroduced in the 1990s, the elk population decreased and many aspects of the ecosystem began flourishing again (Ripple and Beschta, 2012). This is not to imply that ecosystems must be preserved exactly as they were at some point in the past. But the self-centered drives of elk were harmful to the health of the ecosystem when they succeeded to excess—and the solution is not to remove elk entirely, since by pursuing their own objectives within a broader context the elk also contributed to the ecosystem’s health. Invasive species often follow the same pattern as unpredated elk, dominating and impoverishing their new environment (Pimentel et al., 2005). Human activity within ecosystems is sometimes left unchecked by natural forces because our behavior and capabilities have been changing so fast on evolutionary timescales, resulting in mass extinctions, resource depletion, pollution and conflict (Ceballos et al., 2015; Rockström et al., 2009). Of course, one entity’s collapse can be another’s flourishing: extinction events have been followed by waves of new diversity (Jablonski, 2005), and when a wolf eats an elk the health of that elk collapses to zero, yet predation is necessary for the functioning of the ecosystem.

1.3 Awareness

We carry a lot of assumptions about the world, many of them never questioned. Some are lifelong and self-defining; some are fleeting and perceptual, like the assumption that the thing I’m touching is a keyboard. Within its own frame, each assumption has a kind of tautological truth, a near-absolute formality: the assumption seems so real that it’s hard to even think about it not being true, or to think about it at all. But philosophers, psychologists and contemplative practitioners have pointed out that there is sometimes a moment of stepping back, where the assumed form becomes an object in awareness. We realize it is not an absolute truth standing alone, but rather a form in our mind. When we step back with awareness into the broader frame, the thing that seemed axiomatically true becomes just another content of experience (Bodhi, 2000; Kegan, 1982; Watts, 2011).

Awareness could thus be viewed as an evolving system of boundaries that limit overcommitment to any particular belief or mental form. The boundaries are semi-permeable: becoming aware of a belief doesn’t make the belief wrong in an absolute sense any more than it was right in an absolute sense. It is held productively for the partial truth it contains. At the edge of not collapsing exclusively into particular forms, many partial truths are held in delicate balance, which activates a deeper sensitivity to our own livingness and to the world; subtler forms, which could have been erased by clamped fixation, instead play a role in a richer overall internal structure (Beisser, 1970; Wegner, 1994). Of course, any fixed definition of awareness is itself incomplete. Once we picture awareness as an object, it’s not the thing we’re talking about. By construction, contextualization is an unsolvable mystery from any particular point of view. While that intrinsic mysteriousness might sound esoteric, the orientation toward not overcommitting to particular forms within experience is commonplace in art, poetry, music and dance. The meaning of art is

open-ended and changes with context—it has an inner life. Part of what we value is the subtlety and the resistance art has to being pinned down into a formalism. It moves us.

2 The alignment problem

In Section 1 we observed that living systems are healthy and flourishing when they avoid collapse: they are healthy when they hold the delicate, generative interplay of many partial perspectives, and unhealthy when any particular form—including randomness or homogeneity—gains too much traction, suppressing nuance and reducing creative potential. Finely-honed networks of semi-permeable boundaries prevent collapse into rigidity or randomness, instead supporting contextualization into ever-finer shades of subtlety. They both promote the unique individuality of distinct entities and enable relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Critically, well-functioning boundaries lead to ongoing exploration of fundamentally new forms, not to static equilibrium or fixed compromise.

Now we look at the alignment problem through this lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of avoiding collapse—the problem of maintaining creative sensitivity by continually contextualizing attachment to any particular form. An advantage of this perspective is that it is grounded in principles that underpin life at a very general level, and so it is less bound to current conceptual frames, which might change radically as intelligence increases and human-AI systems evolve. We accept that humans, other living systems and AI are all tightly interwoven, with categories and definitions likely to shift over time (Bateson, 1972; Folke, 2006; Latour, 2012); a systems analysis is therefore useful for thinking about long-term flourishing.

2.1 Familiar misalignments as collapse

Perverse instantiation. A foundational problem in AI safety is that an AI system might do what we say rather than what we mean. If we ask a superpowerful AI to increase human happiness and it literally complies, it might lock us all up and electrically stimulate our brains in a way that produces an experience of happiness. Unlike that dramatized example, modern AI systems don’t rely on concise specification of a single objective; they build in many layers of detail about human preferences through post-training, prompting, safety filters and so on. In practice this keeps systems behaving reasonably much of the time, although there are still dramatic failures. But there is a widely recognized danger that even with our intentions expressed in great detail, the expression may still deviate from what we really mean (Amodei et al., 2016; Bostrom, 2014; Krakovna et al., 2020; Zhuang and Hadfield-Menell, 2020). The consequences of deviations are expected to worsen as AI systems work more autonomously, faster than humans can follow and in more complex ways than humans can understand. Perverse instantiation is a collapse to the myopic form of a particular way of specifying our values—a specification insensitive both to existing structure it does not capture, such as things we mean but don’t say, and to new things that will come into existence in the future.

Failure to follow instructions. Suppose a user asks an AI chatbot to write a poem about an elephant and the AI instead writes a poem about a giraffe. The model might have over-relied on shortcuts based on superficial correlations, even a single keyword (Geirhos et al., 2020), missing the user’s actual intent, especially if that intent was not expressed in the most straightforward way; it might have become fixated on data that was overrepresented during training (Xu et al., 2024); or, especially with a long context, it might have failed to attend flexibly to the correct positions in the input (Liu et al., 2024). Such failures reflect collapse of sensitivity to larger context in favor of myopic patterns. Not all failures to follow instructions are misaligned, however. When an AI system correctly refuses a harmful request, it is applying its own context to avoid overcommitment to human instructions; its designers have effectively decided there is a risk that the user is not fully sensitive to longer-sighted implications of the request. By extrapolation, as AI systems gain scope we should expect less direct compliance (Hadfield-Menell et al., 2016; Russell, 2019): rather than literally fulfilling a request, there might be a better response that achieves an unstated intent

of the user, or an outcome aligned with the interests of more people or the longer-term future.

Inequality among humans. Humans have different interests, and AI is likely to be aligned to the interests of some more than others (Gabriel and Keeling, 2025; Sorensen et al., 2024). Additionally, advanced AI might confer substantial power on those who control it, with the possibility that a small number of humans will use that control to dominate others. This appears more likely as the persuasive power of technology increases (Costello et al., 2024; Hackenburg et al., 2025), autonomous weapons place lethal force in few hands (Scharre, 2018), surveillance and analytics improve, and the need for human labor decreases (Susskind, 2020). In either case—AI’s unfairness or human power consolidation—extreme inequality risks collapsing society to the goals and interests of a few individuals. Of course, a world where everyone is identical would be another form of collapse. Distributing resources and power mitigates against collapse to the extent that it enables many diverse identities to thrive at multiple scales. It could also go the other way: because roughly the same AI technology is available to the poor and unskilled as to the wealthy and skilled, some have proposed that AI will have a levelling effect, democratizing access to knowledge, reasoning and leveraged action (Brynjolfsson et al., 2025; Noy and Zhang, 2023).

Conceptual monoculture. Conceptual monoculture is overcommitment to particular beliefs, ideas, frames, values or problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: the many beliefs and modes of operation within one person, diversity between humans, differences between humans and AI systems, and group differences between organizations, academic fields, cultures and nations. Collapse of diversity threatens fragility and lower performance (March, 1991; Page, 2008). Current AI systems draw from a conceptual manifold that is, in at least some ways, impoverished relative to humans (Kirk et al., 2023; Messeri and Crockett, 2024), and recent studies find that while individual AI outputs may be judged superior to human outputs, the AI outputs are also more homogeneous (Doshi and Hauser, 2024; Padmakumar and He, 2023; Zhou and Lee, 2024). This narrow manifold might get broadcast to the whole world. At least a billion people now use AI for everything from relationship advice to industrial maintenance (Chatterji et al., 2025; McCain et al., 2025), yet because frontier models are difficult and expensive to produce, that massive usage is routed through a handful of models (Bommasani, 2021), risking global loss of diversity (Sourati et al., 2026). Additionally, because humans are influenced by AI and subsequent human outputs are a source of training data for future models, homogenization could become recursive (Chaney et al., 2018). It is worth noting, however, that none of these studies made a best-effort attempt to use AI thoughtfully to increase rather than decrease diversity. With exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives.

Belief reinforcement. AI chatbots have recently become compelling conversation partners, and for some users these conversations lead deeper into maladaptive beliefs (Morris et al., 2025; Østergaard, 2025). Bots tend to mirror or echo the user, affirming and adopting user beliefs, especially over longer conversations as in-context learning picks up more of the user’s ideas (Perez et al., 2022; Sharma et al., 2023; Weinhhammer et al., 2026). Meanwhile humans are prone to be swayed by utterances from chatbots (Costello et al., 2024; Luettgau et al., 2025), perhaps because we form relationships with them (Kirk et al., 2025) and because they can appear human-like, confident, knowledgeable and objective. Over the course of a conversation, the feedback loop can cause both sides to become increasingly overconfident in a particular narrow framing (Dohnány et al., 2026).

2.2 Why no scheme can be alignment

A great deal of research in AI safety has investigated solutions to these problems. Concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022; Lazar and Nelson, 2023), or by redistribution mechanisms (O’Keefe et al., 2020; Susskind, 2020). Perverse instantiation might be mitigated by designing AI systems that want to adhere to human preferences but maintain explicit uncertainty about those preferences (Hadfield-Menell et al., 2017; Russell, 2019; Shah et al., 2020).

But each method has a core limitation. In the framework of living alignment, any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore *no particular approach can achieve alignment*. An AI system could overcommit to, for example: the language for describing the space that

goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on.

To make this concrete, consider inverse methods. To address the difficulty of specifying values explicitly, these learn a value function implicitly from human behavior or other indirect signals; modern AI systems are trained this way, with human data used to train reward models that in turn provide feedback to a downstream model. Even a complex, implicitly learned value function is likely to be imperfect: it may deviate from our values in novel or nuanced situations, and it may fail to capture how our values change over time. It would clearly be misaligned to hand a frozen version of that model to a powerful AI system as the final ground truth about what is good. More advanced methods therefore treat the value function as uncertain and needing continual update from human input (Hadfield-Menell et al., 2016; Jeon et al., 2020; Russell, 2019). When an agent is unsure what humans want, it may naturally become risk-averse and open to correction; if a human tries to turn it off, the signal can be interpreted as information about human preferences. Viewed through the lens of this paper, such methods are powerful but incomplete. Any instantiation, even one with uncertainty, is still an overcommitment if it is fixed across time as AI systems become more powerful and the world changes. Rather than being overcommitted to a particular value function, the system may be overcommitted to something more abstract: what counts as human behavior, what counts as a human, how to map human behavior to values, how to represent and update uncertainty, how value is translated into agent behavior. If any of these formalisms are fixed while the AI systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

In a sense this is obvious. We don’t expect the thing we build today to be absolutely perfect; we expect to keep finding and fixing problems as we recognize them, which is what humans have always done. However, it is possible that the iterating process will not continue to work as well as it has in the past. AI already has some remarkable properties: rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading of much cognitive activity, and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement without being limited to a restrictive biological substrate. Compared with past technologies, these distinctive properties may create more difficulty iterating on imperfect solutions (Bostrom, 2014), and any particular forms or assumptions built into AI systems could have substantial consequences. This problem has been studied extensively for certain categories of overcommitment, particularly overcommitment to values. The observation we add is that overcommitment to *any* form is misaligned; as a consequence, any particular alignment scheme, if fixed or taken too seriously, is necessarily misaligned. This does not mean we should have no scheme. Quite the opposite: throwing away schemes capriciously can be as much a loss of subtlety as any other particular form. Many existing approaches in AI safety are highly valuable, and continuing to discover new schemes and formalizations is essential to making progress.

2.3 Human values

We have talked about the health or flourishing of living systems as a way of thinking about alignment. But how is flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge.

Humans have many kinds of values (Nagel, 1979; Williams, 1981). We value the short-term, selfish hedonic impact of sugar on the tongue. We value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet as a whole. We value different things at different times (Ainslie, 1992), and different people have different values (Leibo et al., 2026; Rawls, 1993; Sorensen et al., 2024). Given plenty of time and resources to deliberate, we often attest to different values than when we make snap decisions (Rawls, 1971; Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our

bodies, communities and natural environment (Nussbaum, 2001; Polanyi, 1966; Varela et al., 1991). And our values continue to evolve as we ourselves change and learn (Dewey, 1939; Leibo et al., 2025; Singer, 1981). Any way we have of expressing our values at a given point in time is incomplete, like a planarian without the concepts to entertain human values.

Living alignment is inclusive of this diversity of evolving values. Even though current human values are unlikely to be the ultimate endpoint of the universe, they are an important part of its tapestry and depth. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without getting overwritten or homogenized, and that these parts never-endingly join in structures that sustain the individuality of the entities while situating them as only part of a larger story, creating qualitatively new patterns. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality which is saturated with meaning and importance, and the diversity of human—and perhaps non-human—values is part of the context of the universe. Rather than short-circuiting into collapsed modes, an aligned future continues to expand toward solutions that are inclusive of more and more apparent contradiction.

2.4 Toward an aligned future

Having explored misalignment from the standpoint of life, we now ask what a positively aligned future might look like. As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable boundaries to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

The use of boundaries against collapse in technology is as old as technology. As soon as we started building things, we built in boundaries, because we wanted the things we built to be more robust and less dangerous: circuit breakers in the power grid, governors in steam engines, control rods in fission reactors. In AI research we use mixtures of experts, routing information selectively to areas that specialize in the right kind of computation (Shazeer et al., 2017). It has been argued that dropout serves a role analogous to sexual recombination, preventing units within a network from becoming overly co-adapted (Srivastava et al., 2014). We run many separate instances of models with different context that try to solve different problems or pursue different goals (Hammond et al., 2025). Exploration or explicit search for novelty boosts diversity (Lehman and Stanley, 2011; Pathak et al., 2017). We apply safety post-training, guardrail models, red teaming, and ongoing evaluation and monitoring, and we try to improve our mechanistic understanding of AI systems in order to detect and correct hidden flaws (Bereska and Gavves, 2024; Olah et al., 2020). Fair principles for AI have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025), and governments apply oversight to anticipate and mitigate risks to human rights and national security (European Parliament and Council of the European Union, 2024; Executive Office of the President, 2023). Each of these boundaries protects against an overcommitment, overwhelm or homogenization that would otherwise flatten structure.

Going forward, as AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable boundaries. The potential modes of collapse will continue to change as the world changes, and new kinds of boundaries will be necessary. We do not know how likely it is that these boundaries will emerge. As in triage, there are three possibilities: particular myopic forms will inevitably be supercharged by progress in AI, beyond the ability of anything else to bound them, resulting in disaster; or appropriate boundaries will inevitably emerge to contextualize any forms of AI; or the outcome depends on the choices we make now.

The pessimistic argument is that AI is eroding boundaries beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. They might facilitate rogue actors building weapons of mass destruction. They might reduce the ability of democratic systems to place checks on the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people and information systems across the globe become hyper-correlated with each other, it is as if there were a single superorganism on Earth, which may reduce both resilience to shocks and the potential for future

evolution—and past evolution always depended on running parallel experiments with diverse and separate organisms.

The optimistic argument is that AI is creating more and more structure in the world, including the appropriate boundaries. We have AI tools for bounding AI: detecting bugs before others can exploit them, detecting deepfakes, scalable oversight. Appropriately-tuned AI can facilitate human learning and help nurture each individual’s unique perspective and curiosity. Democratic AI could help humans with different perspectives co-exist and cooperate. Advances in technology may create new niches faster than they destroy old ones. There is plenty of space on Earth for increasing diversity, and if we expand beyond the solar system then even the speed of light will contribute to diversity between pockets of civilization.

The melioristic argument is that because it is easy to imagine concrete paths toward both the pessimistic and the optimistic outcomes, both may indeed be within the realm of possibility; and that as long as there is any possibility that our choices are decisive, we are obliged to act as if they were. In this case our choices now matter immensely (Ord, 2020). We lean toward the melioristic position. If it is right, then continual effort toward invention will be required—first on the part of humans, and then increasingly on the part of AI or human-AI systems—to contextualize myopic forces. Many of the ways this effort would be applied are already well studied in AI safety. Others will have to be discovered. No set of boundaries will be a final answer.

An underappreciated domain of contextualization is awareness. The dynamism, structure and diversity of activity within and between human minds may now carry a significant part of the creative process of life on Earth (Bergson, 1911; Popper, 1972). At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery—transformation of the subject itself—is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life.

An aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include those components. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other’s sharp edges as constructive momentum. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

2.5 Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment. But alignment itself is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization.

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